

**Kingdom of Saudi Arabia
Ministry of Education
Taif University
College of Computers and Information Technology
Department of computer engineer**

Light-Controlled LED Circuit using LDR

1-Introduction

This project presents a simple light-controlled lighting system using a photoresistor (LDR) and a transistor. The aim of the project is to design a circuit that responds to changes in ambient light conditions.

The system is intended to automatically control a light source, making it useful for applications such as automatic lighting in dark environments. The components used in this circuit help demonstrate basic electronic principles, including sensing, signal control, and switching.

2. Equipment and Materials

1. Wires:

Used for making connections between components on the breadboard, enabling circuit functionality and flexibility.



2. 9V Battery:

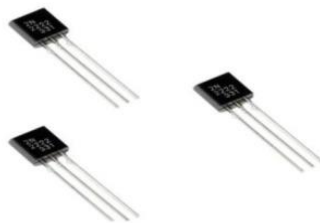
Supplies power to the entire circuit, ensuring all components have the required operating voltage.

**3. 9V Battery Holder:**

Provides a secure and stable connection for the 9V battery to the circuit, allowing easy replacement and portability.

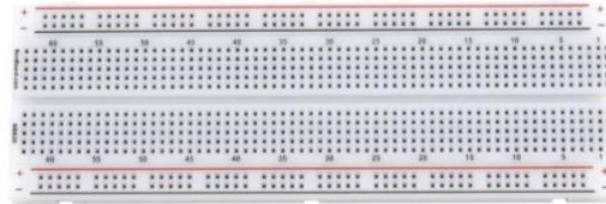
**4. 2N2222 NPN Transistor:**

Small switching transistors used to amplify and control current flow, allowing the Op-amp to control the fan indirectly by switching higher currents.



5. Medium Breadboard:

Serves as the main platform to assemble and organize all components, making it easy to test and modify the circuit.



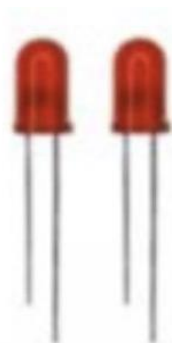
6 . 22K Ohm Resistor:

Used to stabilize current flow within the circuit, helping to balance the Op-amp and thermistor output for consistent performance.



7. LED (Light Emitting Diode)

A Light Emitting Diode (LED) is a semiconductor device that emits light when an electric current passes through it. It operates as a light source in the circuit and only allows current to flow in one direction.



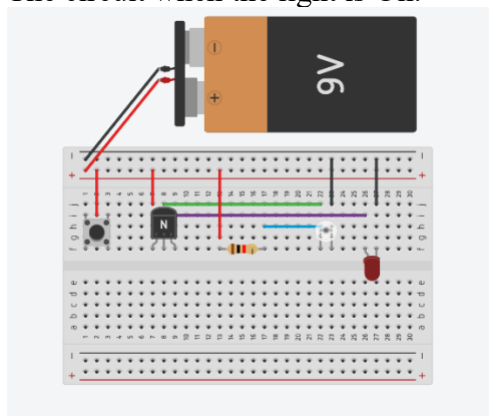
8. Light Dependent Resistor (LDR)

is a device that changes its resistance based on the amount of light it receives. It has a high resistance in the dark and a low resistance in the light. It responds to any color of light.



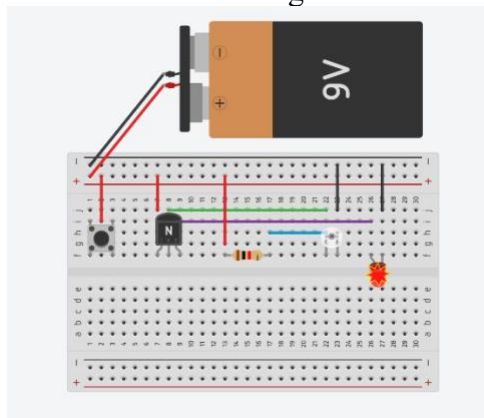
3. Simulation

The circuit when the light is On.



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The circuit when the light is Off.



4. Working Principle

The circuit works by using an LDR (Light Dependent Resistor) to detect changes in the surrounding light intensity. When the room is well-lit, the LDR detects a high light intensity, keeping the LED turned OFF. When the room light is turned OFF, the light intensity decreases, and the LDR detects this change. As a result, the circuit activates the LED automatically.

This allows the LED to respond automatically to changes in the lighting conditions without the need for manual control.

5. Conclusion and Results

The circuit was successfully designed and tested to control an LED according to the surrounding light intensity. When the room light was turned ON, the LED remained OFF. When the room light was turned OFF, the LDR detected the decrease in light intensity and the LED turned ON automatically.

The results show that the circuit can effectively detect changes in light conditions and provide automatic lighting control. This demonstrates a simple practical application of an LDR in an automatic lighting system.